CheckURL

Click with Confidence, We Check First !

Problem

Phishing, corrupted links, and scam websites are currently some of the leading cybersecurity threats people are facing today. Many users may struggle to identify these threats, making them vulnerable to cyberattacks.

Objective

Develop a Firebox web extension that detects and blocks malicious URLs in real time, protecting users from their device being compromised.

System Architecture

(1)
Page Load
Content.js
I/O: User loads page/links go to AI analysis

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(2)
AI analysis
Backround.js
I/O: Links on users page/AI analysis

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(3)
Click Protection
Content.js
I/O: AI analysis/block malicious links

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(4)
Visual Warning
Content.css
Red = Malicious | Yellow = Suspicious | Green = Safe

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(5)
Dashboard
Popup.html
I/O: AI analysis/URL summary on popup

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(6)
Firefox Extension
Manifest.json
I/O: CheckURL folder/Working extension

Solution

CheckURL is a real-time in-browser extension solution. It is designed for Firefox users who are seeking real-time protection against online cyberattacks. It offers a seamless experience, automatically scanning and rating links. With CheckURL's ability to eliminate the need for manually copying and pasting links into external domains, the benefits of using CheckURL are peace of mind and saving your precious time. Users will gain safety verification without having to leave their current tabs. With CheckURL, users will be able to browse the web as normal with an active real-time security system watching their backs.

Key Differences

The key differences between our product and similar competitors is the real time analysis and automation. The most common URL analyzers on the market work by the end-user copying and pasting the link into a website for analysis. CheckURL works differently, after its installed it is constantly searching for malicious links before you click on them.

Impact

There is no software currently out on the market that checks links in real time without having the user manually hover over the link or copying the link and pasting it elsewhere to verify that the link is safe. We aim to have a real time link checker powered by OpenAI to verify links for users to confidently browse the web without fear of online attacks.

Target Customer

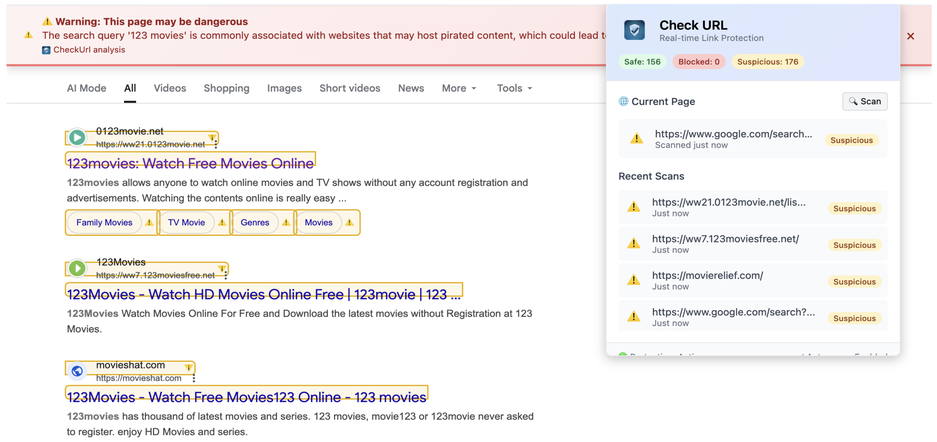
Firefox users who is concerned about the safety of the links they come across online. They spend a lot of time on social media, emails and websites with tons of content where suspicious links show up all the time. These users who want something easy that works automatically an extension which works in real time and protects them.

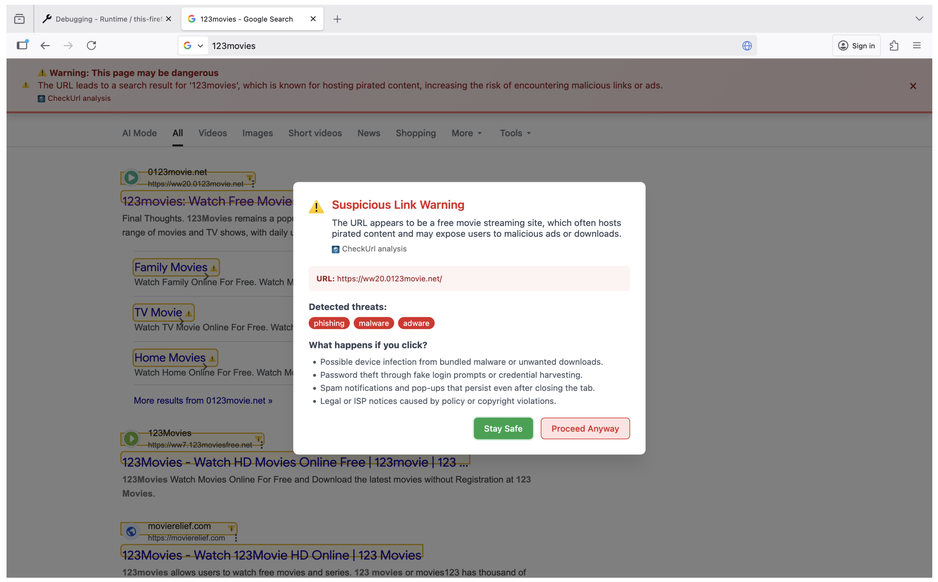
Open AI Prompt

For each link the Open API is give this prompt:

- "You are a cybersecurity expert analyzing URLs for potential threats. Analyze the following URL and determine if it is safe, suspicious, or malicious"

It is told to evaluate the URL based on domain reputations, URL structure, phishing indicators, known malicious patterns, brand impersonation attempts, and social engineering. The API responds with a status (Safe, suspicious, or malicious), the specific threats present, a confidence level and a brief explanation.





Competitor	NordVPN Link Checker	GoDaddy Website Security	Bitdefender Link Checker	VirusTotal	CheckURL
Model	Free tool / Subscription	Subscription	Free tool	Free/Paid API	Browser extension (Firefox only)
Manual or Automatic	Manual/Automatic	Automatic (for websites, not browsing)	Manual	Manual (for users)	Fully automatic
Key Limitation	1. Complex install for Full	Designed for website owners,	Manual check only,	Requires copying/past	Currently niche (Firefox
	VPN suite + multiple settings 2. Intimidating for average users	not individual users	not integrated into browsing experience	ing links, technical for average users	only) but effortless user experience
Real-Time Automatic Scan	No	Yes (for websites only)	No	No	Yes
Built for Everyday Browsers	Yes	No	Yes	Yes	Yes
Simple Visual Alerts	No	No	No	No	Yes
Works Without Copy-Paste	No	No	No	No	Yes
Price	Free for manual $3.89/month	$6.99–$16.99/month	Free	Free	$5.99/month

Referneces

OpenAI. (2025, November 3). *Conversation with ChatGPT about phishing-protection browser extension*. Retrieved from https://chat.openai.com/